The Decision to Hold Back: Teacher and Student Determinants of Grade Repetition

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June 29, 2026

Abstract

Grade repetition is a high-stakes educational decision, but most evidence studies its consequences for students rather than the teachers' decision process that produces it. We conduct a preregistered survey experiment with 2,705 teachers in Castilla-La Mancha, Spain. Teachers evaluate hypothetical student profiles and decide whether each student should repeat a grade. The experiment randomly varies whether teachers are exposed to policy alternatives, whether they rank those policies before assignment, and whether they are explicitly told if the assigned policy matches their preferences. The main result is that teacher and student characteristics explain repetition decisions much more than policy exposure or preference-policy alignment. Teachers respond strongly to student profiles: failed subjects increase repetition by 54.6 percentage points and low competence by 27.1 percentage points within teacher. Harsher teachers place especially large weight on failed subjects. Teacher characteristics and beliefs also matter. Secondary-school teachers, younger teachers, teachers with stronger meritocratic beliefs, and teachers who are more skeptical about resources for repeating students are harsher. In contrast, the preregistered treatment effects are small and mostly indistinguishable from zero. Sensitivity analyses using minimum detectable effects and equivalence tests show that the design can rule out several moderate effects but not all small alignment-specific effects. These results suggest that grade repetition is mainly driven by stable teacher profiles and by the academic information contained in student profiles, while light-touch policy assignment and alignment cues do not substantially alter repetition behavior.

Keywords: grade repetition; teacher decision-making; survey experiment; policy preferences; teacher beliefs.

JEL codes: I21; I24; C93; D91.

1 Introduction

Grade repetition has been regarded as a double punishment for students, who not only face educational failure, but also have to deal with being segregated from their classmates. This can damage students' self-esteem and their perception of what they are capable of achieving. It can also increase costs for the education system and delay educational progression. Despite these concerns, grade repetition remains common in several education systems.

Spain is a particularly relevant setting. In PISA 2022, 22 percent of 15-year-old students in Spain reported having repeated a grade at least once, compared with an OECD average of 9 percent. This difference has made repetition an important education-policy issue. However, most research on grade repetition studies what happens to students after they repeat. Much less is known about the teacher-side decision process that generates repetition recommendations in the first place.

This paper shifts the perspective to teachers' decision-making. We conduct a preregistered survey experiment in Castilla-La Mancha, Spain, with 2,705 teachers. Teachers evaluate eight hypothetical students and decide whether each student should repeat a grade. Each student profile varies across six binary characteristics: gender, migrant or complex background, three or more failed subjects, low mathematical or linguistic competence, absenteeism, and disruptive behavior. This design allows us to estimate which student characteristics matter most for teachers when making repetition decisions.

The experiment also varies teachers' exposure to education policies. The control group completes the repetition task before ranking policies. In the Policy treatment, teachers are randomly assigned to one of three policies before completing the repetition task. In the Revelation treatment, teachers rank the policies before being randomly assigned to one. In the Awareness treatment, teachers rank policies, are assigned to one, and are explicitly informed whether the assigned policy is their favorite, neither favorite nor least favorite, or least favorite policy. This allows us to test whether policy exposure, preference revelation, and preference-policy alignment affect teacher harshness.

The main contribution is not that policies are shown to change teachers' repetition decisions. In fact, they do not appear to do so in a meaningful way. The main contribution is to show that repetition decisions are explained primarily by two sets of factors: the characteristics of the students and the characteristics and beliefs of the teachers. Policies and alignment with policies do not seem to alter this relationship much.

The first set of results characterizes teachers. Following the preregistered analysis plan, we construct several teacher-level harshness measures. The preferred measure compares each teacher's decisions with the decisions of other teachers who saw the same student profile. We then use both ordinary regressions and random forest models to identify which teacher characteristics are associated with greater harshness. Secondary-school teachers are harsher than primary-school teachers. Younger teachers are harsher. Teachers with stronger beliefs in effort and merit are harsher. A narrow resource-skepticism index, based on whether teachers think too many resources are devoted to repeating students and whether those resources are ineffective, is also strongly associated with harshness. In a random forest model, age, teaching experience, time in the current school, contract status, education level, and resource skepticism rank among the most important predictors.

The second set of results characterizes the student profiles. This is the only card-level analysis that we keep in the main text, and we present it as descriptive. Within the same teacher, having three or more failed subjects increases the probability of recommending repetition by 54.6 percentage points. Low mathematical or linguistic competence increases repetition by 27.1 percentage points. Absenteeism and disruptive behavior matter, but much less. Male gender is not relevant, and complex or migrant background is negatively associated with repetition once the other characteristics are held constant. Harsher teachers are not simply more punitive across all dimensions: compared with lenient teachers, they place much greater weight on failed subjects and less weight on behavioral indicators.

The third set of results reports the preregistered hypotheses one by one. Study I compares Policy treatment with Control. Study II compares Revelation treatment with Policy treatment. Study III compares Awareness treatment with Revelation treatment. In each study, we test whether teacher harshness changes with the treatment arm, with the assigned policy, and with preference-policy alignment. The estimates are generally small and statistically insignificant. The only suggestive result is that, in Study II, being assigned to Policy 2 in the Revelation treatment increases harshness relative to being assigned to Policy 2 in the Policy treatment, but this is significant only at the 10 percent level and should be interpreted cautiously.

The final set of results treats minimum detectable effects and equivalence tests as sensitivity analyses. These analyses are useful because many preregistered estimates are null. The question is therefore not only whether the coefficient is statistically significant, but also whether the design can rule out effects that would be meaningful. The answer is mixed. For the broad treatment-arm comparisons, the analysis rules out moderate effects under a 3 percentage point equivalence margin. For more granular policy-assignment and alignment hypotheses, the intervals are wider, so the null results should be interpreted more cautiously.

This paper is related to a growing literature on grade retention, teacher expectations, and teacher decision-making. It is not the first paper to use vignette or survey methods to study teachers' views about repetition. Recent and older work has studied teachers' beliefs about retention and the factors that enter retention judgments. Our contribution is narrower but, we think, important: to our knowledge, this is among the first preregistered survey experiments with teachers that jointly studies repetition decisions, random policy assignment, preference revelation, and explicit preference-policy alignment. This combination allows us to distinguish the determinants of repetition decisions from the effects of policy exposure itself.

2 Related literature

The first related literature studies the effects of grade repetition on students. Repetition is often justified as a way to give students more time to acquire basic competencies, but empirical evidence is cautious. Some studies find short-run improvements in specific contexts, especially when retention is linked to remediation. Other studies find negative medium- or long-run effects on dropout and educational attainment. Alet, Bonnal, and Favard (2013) show that repetition in France produces short-run gains that become negative after several years. Cabrera-Hernandez (2022) studies a reform in Mexico and shows that reducing repetition lowered dropout without reducing average performance. OECD evidence also documents that systems with high repetition rates often do not perform better.

The second literature studies teacher expectations and teacher decisions. Teachers' expectations are consequential because they affect recommendations, grading, tracking, feedback, and student self-perception. This literature emphasizes that teacher judgments may combine valid information, professional experience, and bias. In the context of grade repetition, the relevant question is not simply whether teachers are biased, but which student characteristics they weight when deciding whether a student should repeat.

The third literature studies teachers' beliefs about grade retention. Tomchin and Impara (1992) show that teachers may view repetition as an acceptable practice even when research evidence is skeptical about its benefits. More recent vignette work also shows that academic performance, student behavior, gender, and teachers' own attitudes can affect retention decisions. This is directly connected to our design, because we experimentally vary student characteristics and observe teachers' decisions.

The fourth literature concerns policy exposure and motivated reasoning. Information and policy cues do not always translate into behavioral change, particularly when decisions are anchored in professional routines. In our setting, teachers may have stable views about repetition formed through classroom experience. A short policy assignment or an alignment reminder may therefore be too weak to change the decision rule.

3 Experimental design and data

3.1 The repetition task

Teachers complete a repetition task, also referred to in the preregistration as the card game. Each card represents a hypothetical student with six binary characteristics. The characteristics are: male/female, migrant or complex background/no, failed three or more subjects/no, low mathematical or linguistic abilities/no, serious disciplinary infraction/no, and frequent unjustified absenteeism/no. Teachers are asked whether the student should repeat the grade.

Before the task, teachers are shown two examples. One is an ideal student with no negative characteristics. The other is a student with all characteristics associated with grade repetition. These examples were included to make the task clear and to work as an attention check.

The analysis sample contains 2,705 teachers. Each teacher was asked to evaluate eight students. I do not report the average number of valid teacher-card decisions here because the preregistered design is based on eight cards per teacher and any discrepancy in valid counts should be audited separately.

3.2 Teacher harshness

The preregistration defines several measures of teacher harshness. We keep all of them in the analysis and report the alternatives in the appendix.

The first measure is the number or share of repetitions recommended by the teacher. This is intuitive, but it does not adjust for the fact that some teachers may have seen more difficult profiles.

The second measure, and the preferred one in the main analysis, is the average deviation from other teachers who evaluated the same card. Let R_{ij} be the repetition decision of teacher j for card i, and let $\bar{R}_{i,-j}$ be the repetition rate for the same card among all other teachers. The preferred harshness measure is:

$$H_j^b = \frac{1}{8}\sum_i (R_{ij} - \bar{R}_{i,-j}).$$

A positive value means that the teacher is harsher than other teachers facing the same student profiles.

The third and fourth measures compare the teacher's decision with the predicted probability of repetition for the card. The prediction is obtained from a model using the six student characteristics, and in the more complete version also the education level and school ownership. These alternative measures are useful robustness checks because they adjust for profile difficulty in a parametric way.

3.3 Treatment arms

The experiment has four arms.

Table 1: Experimental design

Arm	Sequence	Interpretation
Control	Repetition task, then policy ranking	Baseline without prior policy exposure
Policy treatment	Random policy assignment, repetition task, then policy ranking	Effect of being exposed to an assigned policy
Revelation treatment	Policy ranking, random policy assignment, repetition task	Effect of revealing preferences before assignment
Awareness treatment	Policy ranking, random assignment, explicit alignment message, repetition task	Effect of being made aware of alignment or disalignment

Notes: The three policies are educational reinforcement in small groups, modified promotion criteria, and teacher training in inclusive methodologies and multi-level classroom management.

3.4 Preregistered hypotheses

The preregistration expresses the hypotheses as null hypotheses. I keep that wording here because it helps interpret the results.

H1|1. Teachers' harshness is not correlated with being assigned to policies. This means that teachers assigned to the Policy treatment are not harder than those from the Control group.

H1|2. Teachers' harshness is not correlated with being assigned to a specific policy. This means that policy assignment has no impact on teachers' harshness.

H1|3. Teachers' harshness is not correlated with the alignment of teachers' preferences and their policy assignment. This means that teachers assigned to their favorite or least favorite policies are neither harder nor softer.

H2|1. Teachers' harshness is not correlated with being asked to rank the policies before policy assignment. This means that teachers assigned to the Revelation treatment are not harder than those from the Policy treatment.

H2|2. Teachers' harshness is not correlated with being assigned to a specific policy after having ranked the policies. This means that the interaction of treatment assignment and policy assignment has no impact on teachers' harshness.

H2|3. Teachers' harshness is not correlated with the alignment of teachers' preferences and their policy assignment when teachers have ranked the policies before assignment. This means that teachers assigned to their favorite or least favorite policies after having ranked them are neither harder nor softer than those who had not revealed them before the ranking.

H3|1. Teachers' harshness is not correlated with acknowledging their opinion of the policy they are being assigned to. This means that teachers assigned to the Awareness treatment are not harder than those from the Revelation treatment.

H3|2. Teachers' harshness is not correlated with being made aware of their opinions when assigning a specific policy. This means that the interaction of treatment assignment and policy allocation has no impact on teachers' harshness.

H3|3. Teachers' harshness is not correlated with the alignment of teachers' preferences and their policy assignment in a setting where they are aware of the alignment or disalignment. This means that teachers are neither harder nor softer when assigned to their favorite or least favorite policies with full awareness of their opinion on that policy.

H4|1. Current decisions are independent of previous decisions in the repetition task. We do not estimate this hypothesis because the analysis data do not contain a validated card-level response order. The numeric card identifiers are design labels rather than observed sequence positions.

H4|2. Policy preferences are orthogonal to policy assignment. This tests conformity in the Policy treatment.

H4|3. Treatment assignment has no effect on policy preferences.

4 Empirical strategy

The confirmatory analysis estimates ordinary least squares models using teacher harshness as the dependent variable. The outcome in the main tables is H_j^b, the average deviation from other teachers seeing the same cards. Study I compares Control and Policy treatment. Study II compares Policy treatment and Revelation treatment. Study III compares Revelation treatment and Awareness treatment. Depending on the hypothesis, the regressions include treatment-arm indicators, assigned-policy indicators, favorite-policy indicators, and interactions between treatment arm and assigned policy.

The descriptive student-profile analysis uses the card-level decisions only to estimate the weight of student characteristics. To avoid confusing teacher harshness with attribute weights, we demean the outcome and the student characteristics within teacher. The model is:

$$Y_{ij} - \bar{Y}_j = \beta_1 X_{ij}^{male} + \beta_2 X_{ij}^{background} + \beta_3 X_{ij}^{failed} + \beta_4 X_{ij}^{competence} + \beta_5 X_{ij}^{absent} + \beta_6 X_{ij}^{disruptive} + \varepsilon_{ij}.$$

This model is descriptive. It answers which characteristics are associated with higher repetition probability within the same teacher.

For teacher characterization, we estimate both OLS regressions and random forest models. The OLS regressions are easy to interpret and imitate the preregistered covariate analysis. The random forest models follow the preregistered plan and are used to rank predictors by permutation importance. Instead of reporting SHAP clouds, we report the mean permutation importance and intervals from repeated forests. We also add the narrow resource-skepticism measure to both the OLS and random forest characterization.

5 Descriptive results

5.1 Teacher characteristics and beliefs

Table 2 characterizes harsher teachers using the preferred harshness measure. The first column includes the preregistered X_1 covariates. The second column adds the X_2 covariates from the second survey. The third column adds the narrow resource-skepticism index.

Table 2: Teacher characteristics associated with harshness

	X1 covariates	X2 covariates	X2 + resource skepticism
Secondary education	0.029***	0.040***	0.034***
	(0.008)	(0.009)	(0.009)
Years teaching	0.000	0.001	0.001
	(0.001)	(0.001)	(0.001)
Years in current school	−0.001	−0.001	−0.001
	(0.001)	(0.001)	(0.001)
Female	−0.007	−0.004	−0.005
	(0.008)	(0.009)	(0.009)
Age	−0.003***	−0.003***	−0.003***
	(0.001)	(0.001)	(0.001)
Public school	0.003	0.001	−0.006
	(0.018)	(0.018)	(0.018)
Temporary contract	0.001	0.000	−0.002
	(0.011)	(0.011)	(0.011)
Number of teaching groups	0.000	0.001	0.001
	(0.001)	(0.001)	(0.001)
Perceived impact on students		0.001	0.002
		(0.003)	(0.003)
Self-reported empathy		−0.001	0.000
		(0.007)	(0.007)
Belief in effort		0.006***	0.005**
		(0.002)	(0.002)
Resource skepticism index			0.021***
			(0.004)
Num.Obs.	2487	2156	2156
R2	0.023	0.031	0.042
R2 Adj.	0.020	0.026	0.037
F	7.412	6.155	7.864

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Several patterns are stable. Secondary-school teachers are harsher than primary-school teachers. Age is negatively associated with harshness: younger teachers are harsher. Belief in effort is positively associated with harshness. Most importantly for the belief analysis, resource skepticism is strongly associated with harshness. A one standard deviation increase in resource skepticism predicts a 2.1 percentage point increase in relative harshness in the characterization regression.

This resource-skepticism index is intentionally narrow. It uses two items: whether too many resources are allocated to repeating students, and whether additional resources for repeating students are ineffective. We do not use the broader belief index in the main paper because it combines too many different concepts and has low internal consistency. The narrow index is more homogeneous and easier to interpret.

Table 3 reports the random forest characterization. This is not a causal ranking, but a prediction-based ranking of which teacher characteristics help explain harshness. Age, teaching experience, time in the current school, contract status, education level, and resource skepticism are among the most important variables. The random forest is not highly predictive overall, but it is useful as a descriptive screening tool.

Table 3: Random forest variable importance for teacher harshness

Variable	Mean importance	Lower interval	Upper interval
Age	0.0022	0.0021	0.0024
Years teaching	0.0022	0.0020	0.0024
Years in current school	0.0018	0.0017	0.0020
Permanent contract	0.0017	0.0016	0.0019
Primary education	0.0014	0.0013	0.0016
Resource skepticism	0.0009	0.0008	0.0010
Perceived impact on students	0.0006	0.0005	0.0008
Number of teaching groups	0.0005	0.0003	0.0006
Public school	0.0003	0.0003	0.0003
Belief in effort	0.0003	0.0001	0.0004
High empathy	0.0001	0.0000	0.0002
Female	-0.0001	-0.0002	-0.0001

Notes: The table reports mean permutation importance across 50 random forests. Intervals summarize variation across repeated forests and should be interpreted as stability intervals, not causal confidence intervals.

5.2 Student characteristics in the repetition task

Table 4 reports the only card-level analysis kept in the main text. The purpose is descriptive: it tells the reader which student characteristics teachers use when deciding repetition.

Table 4: Descriptive within-teacher weights of student characteristics

Attribute	Estimate	Std. Error	95% CI
Male student	0.003	(0.005)	[-0.006, 0.011]
Complex/migrant background	-0.019***	(0.004)	[-0.027, -0.010]
Three or more failed subjects	0.546***	(0.007)	[0.532, 0.560]
Low math/linguistic competence	0.271***	(0.006)	[0.258, 0.283]
Absenteeism	0.064***	(0.004)	[0.055, 0.072]
Disruptive behavior	0.046***	(0.005)	[0.036, 0.055]

Notes: Outcome and student characteristics are demeaned within teacher. Standard errors are clustered by teacher. Coefficients are probability-point effects.

The result is clear. Failed subjects dominate the decision. Holding the other student characteristics constant and comparing cards within the same teacher, having three or more failed subjects increases repetition by 54.6 percentage points. Low competence increases repetition by 27.1 percentage points. Absenteeism and disruptive behavior matter, but much less. Male gender does not matter. Complex or migrant background has a negative coefficient once the other attributes are held constant.

Table 5 shows the same descriptive model by terciles of teacher harshness. This makes the heterogeneity easy to communicate. Harsher teachers place especially high weight on failed subjects. In the lenient tercile, failed subjects increase repetition by 33.2 percentage points. In the middle and harsh terciles, the effect is about 65 percentage points. By contrast, absenteeism and disruptive behavior are not what makes harsh teachers harsher. In the harsh tercile, the coefficient on absenteeism is close to zero and the coefficient on disruptive behavior is negative conditional on the other characteristics. This does not mean that harsh teachers reward disruption. It means that their additional harshness is concentrated in academic failure signals, especially failed subjects.

Table 5: Student-characteristic weights by teacher harshness tercile

Attribute	Lenient tercile	Middle tercile	Harsh tercile
Male student	0.002	0.000	0.009
Complex/migrant background	0.012	-0.039***	-0.038***
Three or more failed subjects	0.332***	0.653***	0.652***
Low math/linguistic competence	0.280***	0.289***	0.250***
Absenteeism	0.087***	0.099***	0.008
Disruptive behavior	0.115***	0.067***	-0.042***

Notes: Each column estimates the within-teacher descriptive model separately for teachers in the corresponding tercile of the preferred harshness measure. Coefficients are probability-point effects. Stars indicate significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

6 Preregistered results

6.1 Study I: Policy treatment vs. Control

Study I compares teachers in the Control group with teachers in the Policy treatment.

H1|1 states that teachers' harshness is not correlated with being assigned to policies. The estimate in Table 6 is small and not significant. Teachers in the Policy treatment are not harsher than teachers in the Control group.

H1|2 states that teachers' harshness is not correlated with being assigned to a specific policy. The assigned-policy coefficients are not statistically significant. This suggests that being assigned to reinforcement, modified promotion criteria, or teacher training does not meaningfully change harshness relative to Control.

Table 6: Study I: H1|1 and H1|2

	H1|1	(2)	H1|2
(Intercept)	0.014	0.014	0.023*
	(0.012)	(0.012)	(0.013)
DPolicy treatment	−0.004		
	(0.013)		
assignedPolicy 1		−0.023	−0.025
		(0.017)	(0.016)
assignedPolicy 2		0.017	0.019
		(0.016)	(0.016)
assignedPolicy 3		−0.007	−0.001
		(0.016)	(0.016)
favoritePolicy 2			0.046**
			(0.020)
favoritePolicy 3			−0.044***
			(0.013)
Num.Obs.	1011	1011	1011
R2	0.000	0.006	0.028
R2 Adj.	−0.001	0.003	0.023
F	0.099	2.034	5.833

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

H1|3 states that teachers' harshness is not correlated with the alignment between their preferences and their policy assignment. Table 7 restricts the sample to Control teachers and treated teachers assigned to either their favorite or least favorite policy. Neither favorite assignment nor least-favorite assignment significantly changes harshness.

Table 7: Study I: H1|3

	(1)	H1|3
(Intercept)	0.014	0.021*
	(0.012)	(0.013)
assignationfavorite	−0.015	−0.011
	(0.016)	(0.016)
assignationleast-favorite	0.005	0.007
	(0.016)	(0.016)
favoritePolicy 2		0.064***
		(0.022)
favoritePolicy 3		−0.044***
		(0.014)
Num.Obs.	769	769
R2	0.002	0.032
R2 Adj.	0.000	0.027
F	0.875	6.293

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

6.2 Study II: Revelation treatment vs. Policy treatment

Study II compares the Revelation treatment with the Policy treatment. The difference is that teachers in Revelation rank policies before policy assignment.

H2|1 states that teachers' harshness is not correlated with being asked to rank the policies before policy assignment. Table 8 shows no significant difference between Revelation and Policy treatment.

H2|2 states that teachers' harshness is not correlated with being assigned to a specific policy after having ranked the policies. The interacted model in Table 8 does not show a broad pattern of treatment-policy effects. The contrast table, Table 9, shows one suggestive result: for Policy 2, Revelation teachers are harsher than Policy-treatment teachers assigned to the same policy. This estimate is significant at the 10 percent level. Since it is one contrast among several and does not survive a more conservative reading, it should be treated as suggestive rather than central.

Table 8: Study II: H2|1 and H2|2

	H2|1	(2)	(3)	H2|2
(Intercept)	0.010	−0.002	0.006	−0.003
	(0.007)	(0.010)	(0.011)	(0.013)
DRevelation treatment	−0.012	−0.012	−0.010	0.007
	(0.010)	(0.010)	(0.010)	(0.017)
assignedPolicy 2		0.022*	0.025**	0.044***
		(0.012)	(0.012)	(0.017)
assignedPolicy 3		0.013	0.018	0.024
		(0.012)	(0.012)	(0.017)
favoritePolicy 2			0.052***	0.053***
			(0.017)	(0.017)
favoritePolicy 3			−0.044***	−0.044***
			(0.010)	(0.010)
DRevelation treatment × assignedPolicy 2				−0.038
				(0.024)
DRevelation treatment × assignedPolicy 3				−0.013
				(0.023)
Num.Obs.	1520	1520	1520	1520
R2	0.001	0.003	0.027	0.028
R2 Adj.	0.000	0.001	0.024	0.024
F	1.570	1.681	8.329	6.325

* p < 0.1, ** p < 0.05, *** p < 0.01

Table 9: Study II: H2|2 contrasts by assigned policy

Assigned policy	Contrast	Estimate	SE	p-value
Policy 1	Policy treatment - Revelation treatment	-0.007	0.017	0.664
Policy 2	Policy treatment - Revelation treatment	0.030*	0.017	0.068
Policy 3	Policy treatment - Revelation treatment	0.006	0.016	0.708

H2|3 states that teachers' harshness is not correlated with preference-policy alignment when preferences have been ranked before assignment. Table 10 estimates separate regressions for teachers assigned to their favorite policy and teachers assigned to their least favorite policy. The Revelation coefficient is not significant in either subsample.

Table 10: Study II: H2|3

	Favorite policy	Least favorite policy
(Intercept)	−0.004	0.040***
	(0.013)	(0.014)
DRevelation treatment	0.013	−0.022
	(0.016)	(0.017)
favoritePolicy 2	0.095***	0.043
	(0.029)	(0.029)
favoritePolicy 3	−0.013	−0.074***
	(0.017)	(0.018)
Num.Obs.	530	483
R2	0.026	0.054
R2 Adj.	0.021	0.048
F	4.718	9.095

* p < 0.1, ** p < 0.05, *** p < 0.01

6.3 Study III: Awareness treatment vs. Revelation treatment

Study III compares Awareness treatment with Revelation treatment. The difference is that Awareness teachers are explicitly told whether the assigned policy is their favorite, neither favorite nor least favorite, or least favorite policy.

H3|1 states that teachers' harshness is not correlated with acknowledging their opinion of the policy they are assigned to. Table 11 shows that Awareness teachers are not significantly harsher than Revelation teachers.

H3|2 states that teachers' harshness is not correlated with being made aware of their opinions when assigning a specific policy. The interacted model does not show significant treatment-policy effects. The contrast table, Table 12, also shows no meaningful differences by assigned policy.

Table 11: Study III: H3|1 and H3|2

	H3|1	(2)	(3)	H3|2
(Intercept)	−0.002	−0.008	0.000	0.002
	(0.007)	(0.010)	(0.011)	(0.013)
DAwareness treatment	−0.011	−0.011	−0.011	−0.014
	(0.010)	(0.010)	(0.010)	(0.017)
assignedPolicy 2		0.010	0.011	0.006
		(0.012)	(0.012)	(0.018)
assignedPolicy 3		0.008	0.011	0.011
		(0.012)	(0.012)	(0.017)
favoritePolicy 2			0.065***	0.065***
			(0.018)	(0.018)
favoritePolicy 3			−0.040***	−0.040***
			(0.010)	(0.011)
DAwareness treatment × assignedPolicy 2				0.010
				(0.024)
DAwareness treatment × assignedPolicy 3				0.000
				(0.024)
Num.Obs.	1535	1535	1535	1535
R2	0.001	0.001	0.024	0.024
R2 Adj.	0.000	−0.001	0.021	0.020
F	1.205	0.660	7.622	5.469

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 12: Study III: H3|2 contrasts by assigned policy

Assigned policy	Contrast	Estimate	SE	p-value
Policy 1	Revelation treatment - Awareness treatment	0.014	0.017	0.423
Policy 2	Revelation treatment - Awareness treatment	0.004	0.017	0.812
Policy 3	Revelation treatment - Awareness treatment	0.014	0.017	0.419

H3|3 states that teachers' harshness is not correlated with preference-policy alignment in a setting where they are aware of the alignment or disalignment. Table 13 shows no significant effect for teachers assigned to their favorite policy or to their least favorite policy.

Table 13: Study III: H3|3

	Favorite policy	Least favorite policy
(Intercept)	0.013	0.003
	(0.014)	(0.015)
DAwareness treatment	−0.022	0.001
	(0.016)	(0.018)
favoritePolicy 2	0.089***	0.060*
	(0.030)	(0.032)
favoritePolicy 3	−0.023	−0.041**
	(0.017)	(0.018)
Num.Obs.	551	497
R2	0.027	0.023
R2 Adj.	0.022	0.017
F	5.076	3.800

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

6.4 Study IV: additional preregistered analyses

H4|1 is not estimated because the analysis data do not contain a validated sequence of card decisions. This is an important limitation. The preregistered test concerns whether current decisions depend on previous decisions, but the available card identifiers should not be interpreted as response order.

H4|2 asks whether policy preferences are orthogonal to policy assignment in the Policy treatment. Appendix Table 17 reports logit models for whether being assigned to a policy predicts later ranking it as favorite or least favorite.

H4|3 asks whether treatment assignment affects policy preferences. Appendix Table 18 reports the share of teachers ranking each policy as favorite or least favorite by treatment arm, with confidence intervals.

7 Sensitivity analyses: MDE and equivalence tests

Because many preregistered estimates are null, we add two sensitivity analyses. The first is the minimum detectable effect with 80 percent power. This tells us which effect sizes the design was able to detect with reasonable probability. The second is an equivalence test. This asks whether we can reject effects outside a given substantive margin.

These tests should not replace the preregistered hypotheses. They help interpret them. For broad treatment-arm contrasts, the MDEs are around 2.7 to 3.8 percentage points. The equivalence tests show that H1|1, H2|1, and H3|1 are equivalent to zero under a 3 percentage pointsmargin. This means that moderate average treatment effects can be ruled out for the main treatment comparisons.

The evidence is weaker for the more granular policy-assignment and alignment hypotheses. Many of those estimates are not equivalent to zero under a 3 percentage pointsmargin. This does not mean they are significant; it means that the design cannot rule out small effects for every policy-specific or alignment-specific contrast. This distinction is important. The paper should state that the policy and alignment estimates are mostly null, while also acknowledging that some granular effects are imprecisely estimated.

8 Discussion

The results point to a simple but important conclusion. Grade repetition decisions are mostly explained by the teachers who make them and the student profiles they evaluate. Policy exposure and preference-policy alignment do not substantially alter repetition behavior.

The descriptive card-level results show that teachers respond strongly to academic information. Failed subjects are by far the strongest predictor of repetition. Low competence also matters substantially. Behavioral indicators matter less. This suggests that repetition decisions are structured around academic failure rather than around a general negative evaluation of the student.

The teacher-characterization results show that harshness is also related to teacher profiles and beliefs. Secondary-school teachers are harsher, younger teachers are harsher, and teachers with stronger beliefs in effort and resource skepticism are harsher. These patterns are not causal, but they are informative for policy. If policymakers want to understand why repetition remains common, it is not enough to look at policy rules. They also need to understand the professional beliefs and contexts that shape teachers' thresholds.

The preregistered treatment results are mostly null. Teachers do not become substantially harsher or softer when exposed to policy alternatives, when they reveal preferences before assignment, or when they are made aware that the assigned policy matches or conflicts with their preferences. This is particularly relevant for the alignment hypotheses. The experiment was designed to test whether teachers might respond differently when they feel heard or ignored. We find little evidence that this mechanism changes repetition decisions.

The paper therefore should not be framed as a story in which preferences move but behavior does not. That may be an interesting secondary observation, but it is not the main contribution. The stronger contribution is that student characteristics and teacher characteristics explain repetition decisions, while policies and alignment do not appear to alter them much.

9 Conclusion

This paper studies teacher decision-making in grade repetition using a preregistered survey experiment with 2,705 teachers. Teachers evaluated hypothetical student profiles and were randomly assigned to policy environments that varied policy exposure, preference revelation, and awareness of preference-policy alignment.

The main finding is that repetition decisions are stable with respect to these policy manipulations. The preregistered treatment effects are generally small and statistically insignificant. Sensitivity analyses rule out several moderate average effects, although not all small granular effects. At the same time, repetition decisions are far from random. Teachers respond strongly to student characteristics, especially failed subjects and low competence. Harsher teachers place particularly strong weight on failed subjects. Teacher characteristics and beliefs, including resource skepticism, are associated with higher harshness.

The policy implication is that light-touch exposure to policies or alignment messages is unlikely to change repetition behavior on its own. If policymakers aim to reduce unnecessary repetition, they may need to focus on decision protocols, credible remediation alternatives, and the beliefs that shape teachers' thresholds for repetition.

A Alternative harshness measures

The preregistration proposed several harshness measures. Table 14 repeats the characterization regression using four outcomes: number/share of repetitions, average deviation from other teachers seeing the same cards, deviation from predicted repetition probability, and deviation from predicted repetition probability in a more complete model.

Table 14: Teacher characterization using alternative harshness measures

	Number repetitions	Average deviation	Predicted deviation	Complete predicted deviation
Secondary education	0.035***	0.034***	0.033***	0.002
	(0.009)	(0.009)	(0.009)	(0.009)
Years teaching	0.001	0.001	0.001	0.001
	(0.001)	(0.001)	(0.001)	(0.001)
Years in current school	−0.001	−0.001	−0.001	−0.001
	(0.001)	(0.001)	(0.001)	(0.001)
Female	−0.002	−0.005	−0.005	−0.005
	(0.009)	(0.009)	(0.009)	(0.009)
Age	−0.003***	−0.003***	−0.003***	−0.003***
	(0.001)	(0.001)	(0.001)	(0.001)
Public school	−0.008	−0.006	−0.006	−0.007
	(0.019)	(0.018)	(0.018)	(0.018)
Temporary contract	−0.001	−0.002	−0.002	−0.002
	(0.012)	(0.011)	(0.011)	(0.011)
Number of teaching groups	0.001	0.001	0.001	0.001
	(0.001)	(0.001)	(0.001)	(0.001)
Perceived impact on students	0.003	0.002	0.002	0.002
	(0.003)	(0.003)	(0.002)	(0.002)
Self-reported empathy	−0.001	0.000	0.000	0.000
	(0.007)	(0.007)	(0.007)	(0.007)
Belief in effort	0.005**	0.005**	0.005**	0.005**
	(0.002)	(0.002)	(0.002)	(0.002)
Resource skepticism index	0.021***	0.021***	0.021***	0.021***
	(0.004)	(0.004)	(0.004)	(0.004)
Num.Obs.	2156	2156	2156	2156
R2	0.041	0.042	0.042	0.033
R2 Adj.	0.036	0.037	0.037	0.028
F	7.615	7.864	7.877	6.189

* p < 0.1, ** p < 0.05, *** p < 0.01

B MDE and equivalence tests

Table 15 reports the minimum detectable effects for the preregistered teacher-level hypotheses. Table 16 reports equivalence tests using a 3 percentage point margin. The script also exports the same equivalence analysis using 2 and 5 percentage point margins.

Table 15: Minimum detectable effects for preregistered teacher-level hypotheses

Hypothesis	Estimate	Std. Error	95% CI	MDE 80%
H1|1: Policy treatment vs Control	-0.004	(0.013)	[-0.031, 0.022]	0.038
H1|2: assigned Policy 1 vs Control	-0.025	(0.016)	[-0.057, 0.007]	0.046
H1|2: assigned Policy 2 vs Control	0.019	(0.016)	[-0.013, 0.051]	0.046
H1|2: assigned Policy 3 vs Control	-0.001	(0.016)	[-0.033, 0.031]	0.045
H1|3: assigned favorite vs Control	-0.011	(0.016)	[-0.042, 0.020]	0.044
H1|3: assigned least favorite vs Control	0.007	(0.016)	[-0.025, 0.038]	0.045
H2|1: Revelation vs Policy	-0.012	(0.010)	[-0.031, 0.007]	0.027
H2|2: Revelation vs Policy: Policy 1	-0.007	(0.017)	[-0.041, 0.026]	0.047
H2|2: Revelation vs Policy: Policy 2	0.030*	(0.017)	[-0.002, 0.063]	0.047
H2|2: Revelation vs Policy: Policy 3	0.006	(0.016)	[-0.026, 0.038]	0.045
H2|3: favorite assigned, Revelation vs Policy	0.013	(0.016)	[-0.018, 0.044]	0.044
H2|3: least favorite assigned, Revelation vs Policy	-0.022	(0.017)	[-0.054, 0.011]	0.046
H3|1: Awareness vs Revelation	-0.011	(0.010)	[-0.031, 0.009]	0.028
H3|2: Awareness vs Revelation: Policy 1	0.014	(0.017)	[-0.020, 0.048]	0.049
H3|2: Awareness vs Revelation: Policy 2	0.004	(0.017)	[-0.030, 0.038]	0.048
H3|2: Awareness vs Revelation: Policy 3	0.014	(0.017)	[-0.020, 0.047]	0.048
H3|3: favorite assigned, Awareness vs Revelation	-0.022	(0.016)	[-0.054, 0.009]	0.045
H3|3: least favorite assigned, Awareness vs Revelation	0.001	(0.018)	[-0.033, 0.036]	0.049

Table 16: Equivalence tests for preregistered teacher-level hypotheses, 3 percentage point margin

Hypothesis	Margin	Estimate	TOST p	Equivalent
H1|1: Policy treatment vs Control	+/-0.03	-0.004	0.028	Yes
H1|2: assigned Policy 1 vs Control	+/-0.03	-0.025	0.376	No
H1|2: assigned Policy 2 vs Control	+/-0.03	0.019	0.243	No
H1|2: assigned Policy 3 vs Control	+/-0.03	-0.001	0.036	Yes
H1|3: assigned favorite vs Control	+/-0.03	-0.011	0.109	No
H1|3: assigned least favorite vs Control	+/-0.03	0.007	0.073	No
H2|1: Revelation vs Policy	+/-0.03	-0.012	0.033	Yes
H2|2: Revelation vs Policy: Policy 1	+/-0.03	-0.007	0.091	No
H2|2: Revelation vs Policy: Policy 2	+/-0.03	0.030	0.511	No
H2|2: Revelation vs Policy: Policy 3	+/-0.03	0.006	0.070	No
H2|3: favorite assigned, Revelation vs Policy	+/-0.03	0.013	0.137	No
H2|3: least favorite assigned, Revelation vs Policy	+/-0.03	-0.022	0.313	No
H3|1: Awareness vs Revelation	+/-0.03	-0.011	0.029	Yes
H3|2: Awareness vs Revelation: Policy 1	+/-0.03	0.014	0.176	No
H3|2: Awareness vs Revelation: Policy 2	+/-0.03	0.004	0.065	No
H3|2: Awareness vs Revelation: Policy 3	+/-0.03	0.014	0.173	No
H3|3: favorite assigned, Awareness vs Revelation	+/-0.03	-0.022	0.320	No
H3|3: least favorite assigned, Awareness vs Revelation	+/-0.03	0.001	0.052	No

C Additional preregistered analyses

Table 17: H4|2: Policy assignment and later policy preferences

	Policy 1 favorite	Policy 1 least favorite	Policy 2 favorite	Policy 2 least favorite	Policy 3 favorite	Policy 3 least favorite
Assigned to policy	0.109	0.469	−0.002	−0.099	0.424***	−0.372*
	(0.156)	(0.330)	(0.258)	(0.174)	(0.158)	(0.197)
Num.Obs.	761	761	761	761	761	761
F	0.485	2.022	0.000	0.325	7.169	3.568

* p < 0.1, ** p < 0.05, *** p < 0.01

Table 18: H4|3: Policy preferences by treatment arm

Preference	Treatment	Policy	Share	95% CI
Favorite	Control	Policy 1	58.4%	[52.3, 64.5]
Favorite	Control	Policy 2	10.8%	[7.0, 14.6]
Favorite	Control	Policy 3	30.8%	[25.1, 36.5]
Favorite	Policy treatment	Policy 1	55.3%	[51.8, 58.9]
Favorite	Policy treatment	Policy 2	9.9%	[7.7, 12.0]
Favorite	Policy treatment	Policy 3	34.8%	[31.4, 38.2]
Favorite	Revelation treatment	Policy 1	51.6%	[48.1, 55.2]
Favorite	Revelation treatment	Policy 2	9.1%	[7.0, 11.1]
Favorite	Revelation treatment	Policy 3	39.3%	[35.8, 42.7]
Favorite	Awareness treatment	Policy 1	53.7%	[50.2, 57.2]
Favorite	Awareness treatment	Policy 2	8.1%	[6.2, 10.0]
Favorite	Awareness treatment	Policy 3	38.1%	[34.7, 41.6]
Least favorite	Control	Policy 1	9.2%	[5.6, 12.8]
Least favorite	Control	Policy 2	72.0%	[66.4, 77.6]
Least favorite	Control	Policy 3	18.8%	[14.0, 23.6]
Least favorite	Policy treatment	Policy 1	5.3%	[3.7, 6.8]
Least favorite	Policy treatment	Policy 2	74.1%	[71.0, 77.2]
Least favorite	Policy treatment	Policy 3	20.6%	[17.8, 23.5]
Least favorite	Revelation treatment	Policy 1	6.9%	[5.1, 8.6]
Least favorite	Revelation treatment	Policy 2	75.9%	[72.8, 78.9]
Least favorite	Revelation treatment	Policy 3	17.3%	[14.6, 19.9]
Least favorite	Awareness treatment	Policy 1	4.9%	[3.4, 6.4]
Least favorite	Awareness treatment	Policy 2	78.4%	[75.5, 81.2]
Least favorite	Awareness treatment	Policy 3	16.8%	[14.1, 19.4]

D Card-level robustness checks

The main text keeps only the descriptive card-level attribute analysis. Additional card-level models, including treatment-by-attribute checks and belief-by-attribute checks, are generated by `2. Code/4. decision_card_models.R` and should be read as robustness or exploratory appendix material rather than as central results.

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